

ARFE

Accountable Remote Forensic Examination

What we want to integrate into UpDesk — and whether any remote-access tool already has it

The question this document answers

We want to turn UpDesk (a peer-to-peer remote desktop) into a tool that produces **court-admissible, tamper-evident evidence of a live remote session** — proving exactly what a remote investigator saw and did on a machine. The obvious question before building: **does any existing remote-access or forensic tool already do this?** Short answer: **parts of it exist in four different product categories, but no single tool combines them — and the specific combination is an open gap.**

What ARFE integrates, in one sentence

A live interactive remote-examination tool that produces a **tamper-evident, dual-signed record binding every examiner action to the exact system state-change it caused** — so a court can cryptographically separate what the examiner did from what was already there on the machine.

The five things we integrate

- **Dual-signed session sealing** — every screen frame + input event is signed by BOTH the examiner's key and the target machine's device key (UpDesk already has Ed25519 keys on both ends).
- **Hash-chained tamper-evidence** — the whole session is a Merkle/hash chain; any later edit, insertion or reordering becomes mathematically detectable.
- **Trusted timestamping + public anchoring** — RFC 3161 timestamp and a public transparency log, so the record is verifiable by anyone WITHOUT trusting a vendor.
- **Causal action-to-state attribution** — bind “this click → this file/registry change,” separating examiner-induced changes from original evidence.
- **Court-ready certification** — auto-generate a chain-of-custody + India BSA 2023 Section 63(4) evidence certificate from the sealed session.

How it fits onto UpDesk

ARFE is a layer on top of the existing UpDesk stack. The examiner (controller) and the target machine (native host) already talk over a direct, end-to-end encrypted WebRTC channel with Ed25519 identities on both ends. ARFE inserts an **evidence sealer** that captures every frame and input event, hash-chains and dual-signs them, timestamps and anchors the record, and lets an **offline verifier** (or a court) check it later without trusting the server or any vendor.

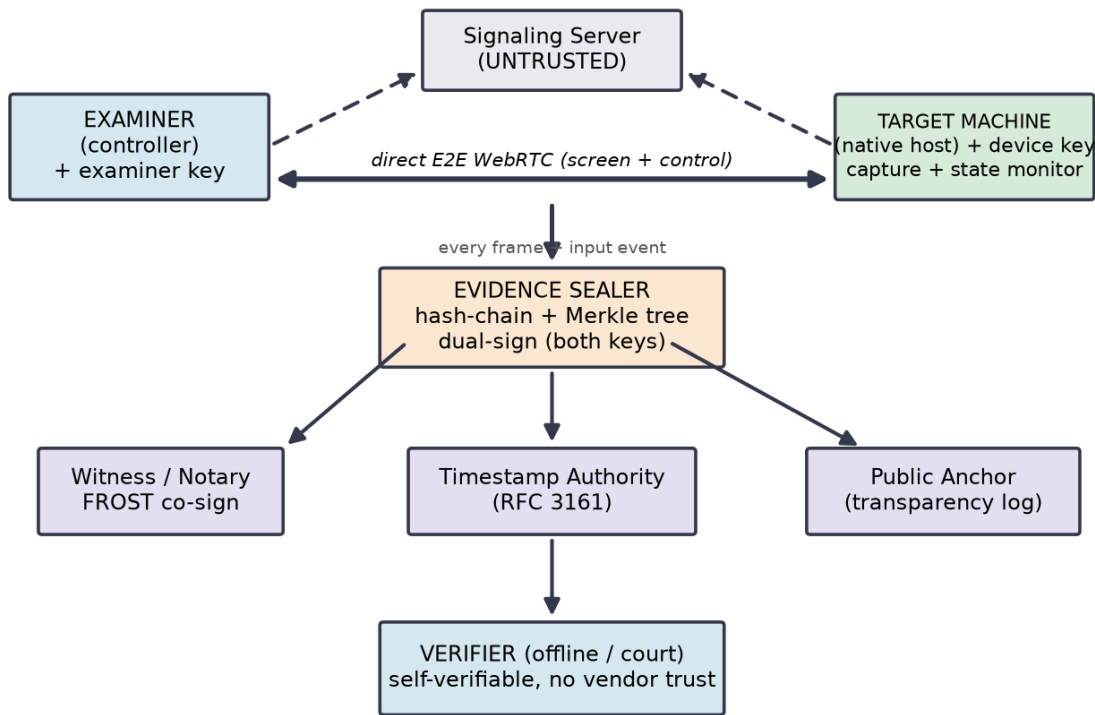


Figure 1: ARFE architecture. Purple = the forensic layer added on top of UpDesk's existing E2E remote-desktop path. The signaling server stays untrusted; the verifier needs no vendor trust.

Why UpDesk is uniquely suited (and most tools are not)

Forensic accountability needs BOTH endpoints to have a cryptographic identity. UpDesk already gives every controller and host an Ed25519 key — so the dual-signature, the part competitors would have to build from scratch, is **already there**. That is the reason this pivot is realistic on this codebase specifically.

Does any remote-access tool already have this?

We checked the real product market across four categories. Each owns a PIECE of ARFE. None owns the whole thing. Here is the honest map:

Capability	Remote desktop (TeamViewer / AnyDesk / RustDesk)	Forensic collectors (GRR / Velociraptor / EnCase)	Forensic capture (TrueScreen / ProofSnap)	PAM session rec. (CyberArk / BeyondTrust)	UpDesk + ARFE
Interactive screen + control	YES	no (headless)	no	YES	YES
Session recording	some	no	static / video	YES	YES
Cryptographic tamper-evidence	no	artifacts only	YES	vault-based	YES
Self-verifiable offline (no vendor trust needed)	no	partial	YES	NO	YES
Dual device-bound signatures (operator + target)	no	no	no	no	YES
Causal input→state attribution	no	no	no	no	YES
Court admissibility (ISO 27037 / BSA Sec 63)	no	YES	YES	no	YES
Works on an untrusted / suspect machine	no	YES	no	no	YES (goal)
Self-hosted / E2E peer-to-peer	RustDesk only	YES	no	no	YES

Table 1: Green = has it, Red = lacks it. The last column (UpDesk + ARFE) is the only one with every row — that combination is the gap.

Verdict: No single product has all of ARFE built in

The individual ingredients exist and are mature — we are not reinventing them. But the specific combination (interactive remote examination + self-verifiable dual-signed tamper-evidence + causal attribution + court certification, on a possibly-untrusted machine) is an **empty cell** in the market.

The closest competitor — and why it still isn't ARFE

The nearest thing on the market is **Privileged Access Management (PAM)** — CyberArk PSM and BeyondTrust PASM. They already do interactive privileged-session recording with cryptographic tamper-resistance, stored for forensic review. That is a real chunk of ARFE, so we have to be precise about the difference.

PAM cannot simply add these — they break its architecture

- 1. PAM trusts a central vault.** Its integrity claim is “stored in our tamper-proof vault.” ARFE's dual device-bound signatures make the record verifiable by a court **WITHOUT** trusting any vendor. PAM cannot be self-verifiable without abandoning the vault model.
- 2. PAM manages your OWN fleet.** It assumes machines you own, with an agent and vaulted credentials. It cannot examine a suspect's machine you do not control.
- 3. PAM is built for compliance / insider audit, not court.** No ISO 27037 or India Section 63 certificate, no external chain of custody.
- 4. PAM records what happened, but has no causal binding** of “this action caused this state change.”

So ARFE's distinct pitch is a single sentence: **self-verifiable, court-admissible, works on a suspect's machine, with no vendor to trust.** That is what PAM structurally cannot say.

The legal hook that makes this timely (India)

India's new evidence law asks for exactly what ARFE provides

The **Bharatiya Sakshya Adhinyam (BSA) 2023** replaced the Indian Evidence Act on 1 July 2024. Its **Section 63(4)** now requires **DUAL certification of electronic evidence — by the person in charge of the device AND an independent expert.** ARFE's dual-signature design (target-device key + examiner key) is the cryptographic embodiment of that exact legal requirement. The design isn't just novel — it answers a brand-new statutory demand.

How we build it — and how long

ARFE sits on top of UpDesk's existing Ed25519 keys, capture pipeline and WebRTC channel, so the foundation is already in place. It is built in phases, each independently useful:

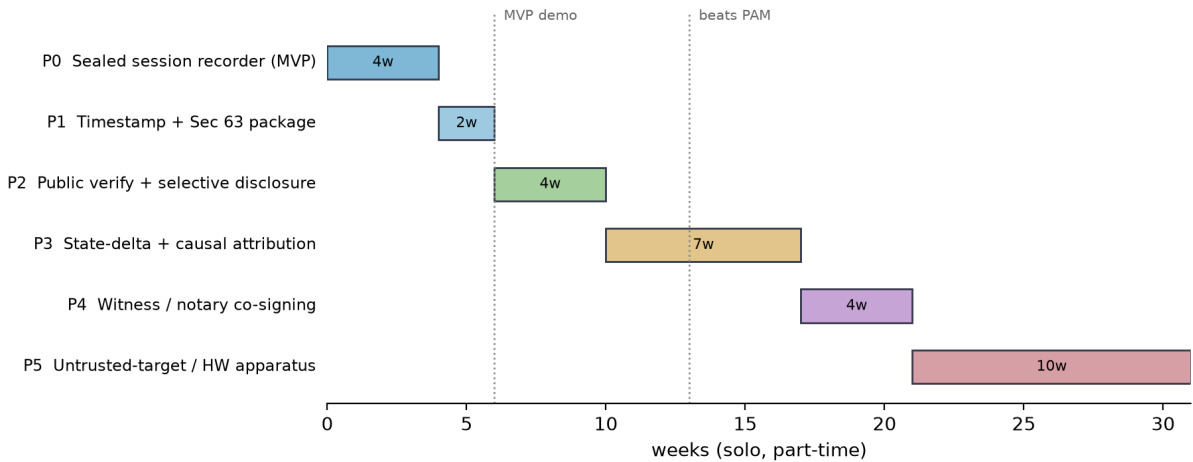


Figure 2: Phased roadmap (solo, part-time, in weeks). The features that most differentiate ARFE from PAM (public verifiability, witness co-signing, causal attribution) are the harder second half.

- **Working demo (Phases 0–1): ~4–6 weeks** — sealed dual-signed sessions + a Section 63 evidence package. Enough to demo and to anchor a paper.
- **Prototype that genuinely beats PAM (Phases 0–2 + partial 4): ~2–3 months.**
- **Full thesis-grade system (through Phase 4): ~5–6 months** — a semester / internship.
- **Hardware-isolated capture on an untrusted target (Phase 5): additional months** — research-grade, and the only angle with a genuine (still uncertain) patent path.

Honest caveats

The building blocks (FROST threshold signatures, RFC 3161 timestamping, Merkle proofs, kernel provenance) all exist as open-source — good for building, but it means the **software** is publishable, not patentable. Phases 3 and 5 carry real research risk (may only partially work — fine for a thesis). A patent, if pursued at all, lives only in the Phase 5 hardware apparatus and would need a professional prior-art search first.